Hence by an application of the l'Hôpital Rule we get,

$$\lim_{x\to a}\left(\frac{x^n-a^n}{x-a}\right) = \lim_{x\to a}\left(\frac{n\times x^{n-1}}{1}\right)$$

$$= n\times a^{n-1}.$$

Example 7.35

Evaluate the limit $\lim_{x\to 0}\left(\frac{\sin mx}{x}\right)$.

Solution

If we directly substitute $x=0$ we get an indeterminate form $\frac{0}{0}$ and hence we apply the l'Hôpital's rule to evaluate the limit as,

$$\lim_{x\to 0}\left(\frac{\sin mx}{x}\right) = \lim_{x\to 0}\left(\frac{m\times\cos mx}{1}\right)$$

$$= m$$

The next example tells that the limit does not exist.

Example 7.36

Evaluate the limit $\lim_{x\to 0}\left(\frac{\sin x}{x^2}\right)$.

Solution

If we directly substitute $x=0$ we get an indeterminate form $\frac{0}{0}$ and hence we apply the l'Hôpital's rule to evaluate the limit as,

$$\lim_{x\to 0^+}\left(\frac{\sin x}{x^2}\right) = \lim_{x\to 0^+}\left(\frac{\cos x}{2x}\right) = \infty$$

$$\lim_{x\to 0^-}\left(\frac{\sin x}{x^2}\right) = \lim_{x\to 0^-}\left(\frac{\cos x}{2x}\right) = -\infty$$

As the left limit and the right limit are not the same we conclude that the limit does not exist.

Remark

One may be tempted to use the l'Hôpital's rule once again in $\lim_{x\to 0^+}\left(\frac{\cos x}{2x}\right)$ to conclude

$$\lim_{x\to 0^+}\left(\frac{\cos x}{2x}\right) = \lim_{x\to 0^+}\left(\frac{-\sin x}{2}\right) = 0.$$

which is not true because it was not an indeterminate form.

Example 7.37

If $\lim_{\theta\to 0}\left(\frac{1-\cos m\theta}{1-\cos n\theta}\right)=1$, then prove that $m=\pm n$.

Solution

As this is an indeterminate form $\left(\frac{0}{0}\right)$, using the l'Hôpital's Rule

$$\lim_{\theta\to 0}\left(\frac{1-\cos m\theta}{1-\cos n\theta}\right) = \lim_{\theta\to 0}\left(\frac{m\sin m\theta}{n\sin n\theta}\right)$$

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